

# WeatherWise API Documentation

AI-Powered Weather Information & Recommendation Service (ai-weatherwise-api v1.0.0)

|                                                |                                       |
|------------------------------------------------|---------------------------------------|
| Project Name: WeatherWise (ai-weatherwise-api) | Team ID: SWTID-2026-9692              |
| Date: 29 September 2026                        | Server Port: 5001 (MongoDB Connected) |

## 1. Executive Summary & System Overview

**WeatherWise** is a backend RESTful API service built with Node.js, Express, and MongoDB. It provides user authentication, real-time weather retrieval for cities (such as Chennai), weather summary generation, and AI-driven personalized weather recommendations.

## 2. API Endpoints & Request/Response Specification

| Method | Endpoint                       | Sample Payload / Parameters                                                   | Response Output                                                                              |
|--------|--------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| POST   | /api/auth/register             | <pre>{ "name": "shree", "email": "shree@gmail.com", "password": "..."} </pre> | <pre>{ "success": true, "data": { "_id": "6ab6013b...", "token": "eyJ..." } } </pre>         |
| POST   | /api/auth/login                | <pre>{ "email": "shree@gmail.com", "password": "..."} </pre>                  | <pre>{ "success": true, "data": { "_id": "6ab6013b...", "token": "eyJ..." } } </pre>         |
| GET    | /api/weather/chennai           | N/A (URL Path parameter)                                                      | <pre>{ "success": true, "data": { "city": "Chennai", "temp": 32.8, "humidity": 63 } } </pre> |
| POST   | /api/ai/weather-summary        | <pre>{ "city": "Chennai", "temperature": 32.8, "humidity": 63, ... } </pre>   | <pre>{ "success": true, "summary": "Today's weather in Chennai is warm..." } </pre>          |
| POST   | /api/ai/weather-recommendation | <pre>{ "temperature": 32.8, "condition": "Clouds" } </pre>                    | <pre>{ "success": true, "recommendation": "Stay hydrated, wear light cotton..." } </pre>     |

## 3. Communication Plan

| S.No | Communication Type     | Frequency | Channel / Tool           | Participants       | Purpose                                 |
|------|------------------------|-----------|--------------------------|--------------------|-----------------------------------------|
| 1    | Team Standup           | Daily     | Discord / Slack          | Core Dev Team      | Sync backend integration progress       |
| 2    | API Testing & Review   | Weekly    | Thunder Client / Postman | Dev Team & QA      | Verify API endpoints & responses        |
| 3    | Issue / Bug Discussion | As Needed | GitHub / Terminal Logs   | Backend Developers | Debug API errors and integration issues |

## 4. Technical Challenges & Resolutions

| S.No | Challenge Faced                                                           | Resolution / Action Taken                                                                                                |
|------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 1    | Gemini API model deprecation error when calling weather summary endpoint. | Updated the model string from <code>gemini-2.5-flash</code> to <code>gemini-3.8-flash</code> per updated API guidelines. |
| 2    | Ensuring secure authentication across API requests.                       | Implemented JWT (JSON Web Token) based user registration and authorization routes.                                       |